Bank Customer
Churn Analysis

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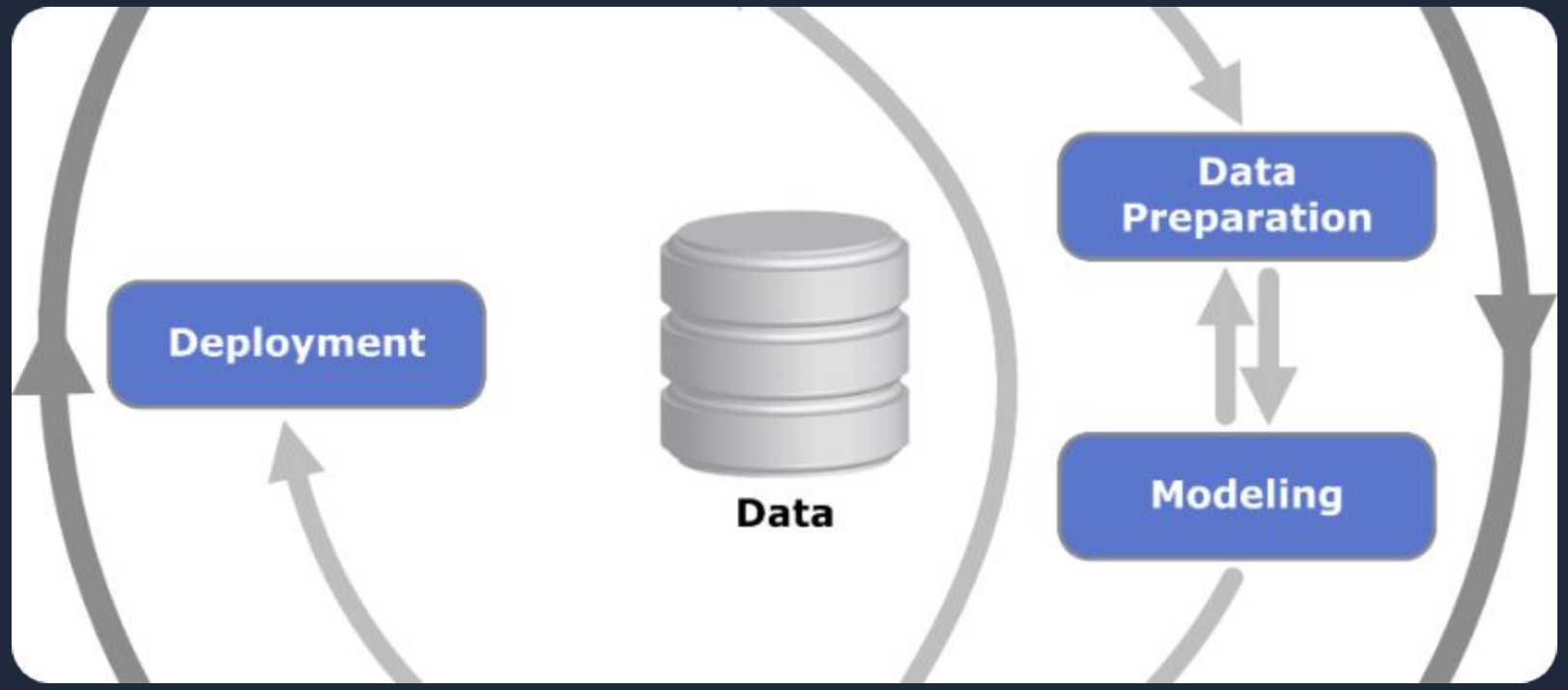
The CRISP-DM Framework

Our data-driven journey from raw numbers to boardroom execution.



EXECUTIVE SUMMARY

20%

AGGREGATE CHURN RATE

2,000 out of 10,000 customers exited during the analysis period.

Critical Takeaways

- **75%** of churners leave with active capital.
- **Germany** faces double the churn risk.
- **Product Bundling** beyond 2 items is lethal.

BUSINESS UNDERSTANDING

Problem Statement

The bank is experiencing a significant capital outflow. While 20% churn is high, the *quality* of customers leaving is even more concerning—financially stable, long-term clients are migrating to competitors.

Target 1

Identify churn drivers across demographics.

Target 2

Quantify financial impact of attrition.

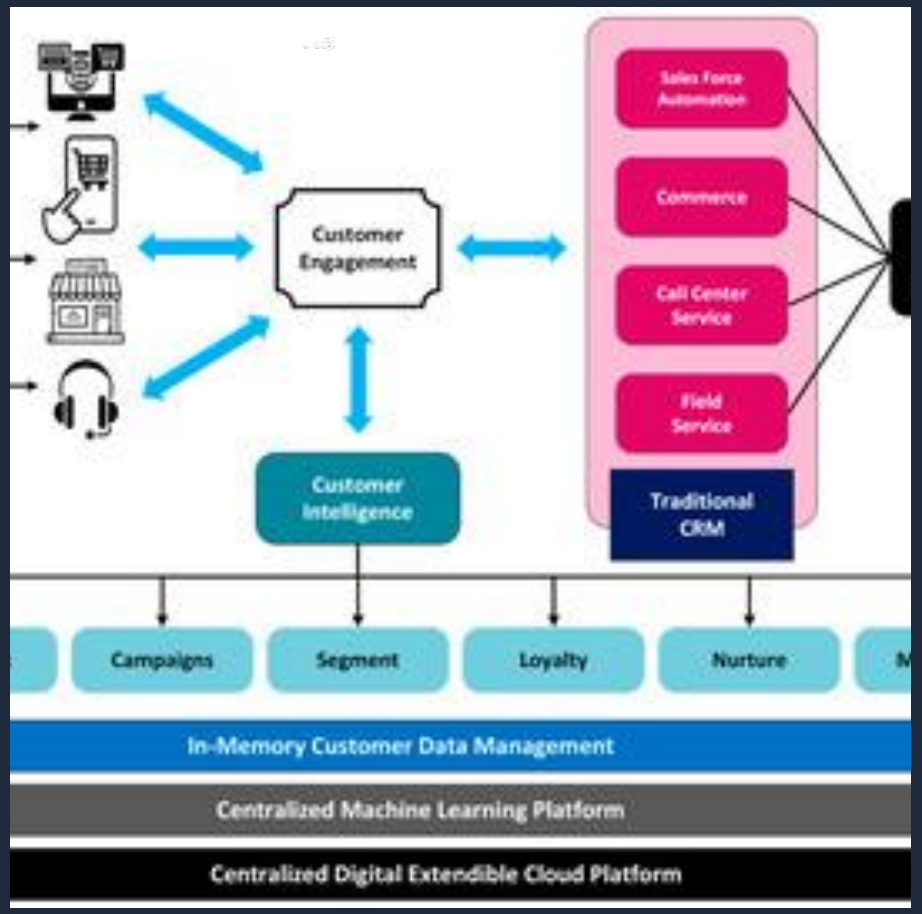
Target 3

Optimize retention budget allocation.

Target 4

Re-engineer product offering strategy.

DATA UNDERSTANDING



Dataset Profile

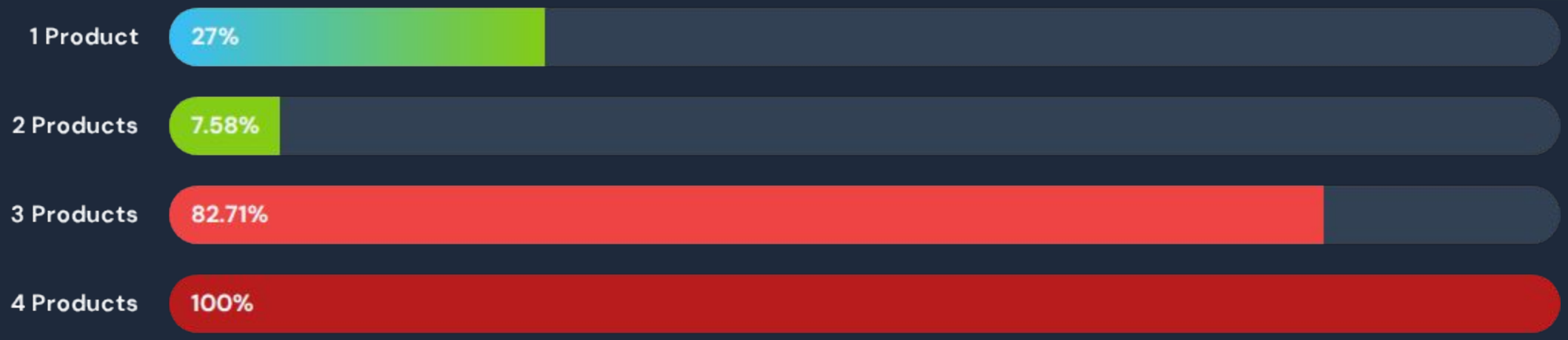
We analyzed **10,000 unique records** spanning demographics, product usage, and credit history.

Temporal scope: 5-year average tenure.

Variables: Credit Score, Balance, Gender, Country.

Integrity: No missing or duplicate values found.

THE PRODUCT CROSS-SELLING ANOMALY



Strategic Pivot: Cross-selling to a 2nd product **maximizes loyalty**. However, exceeding 2 products triggers a total breakdown in retention—likely due to service friction or hidden fees.

GEOGRAPHIC HOTSPOTS

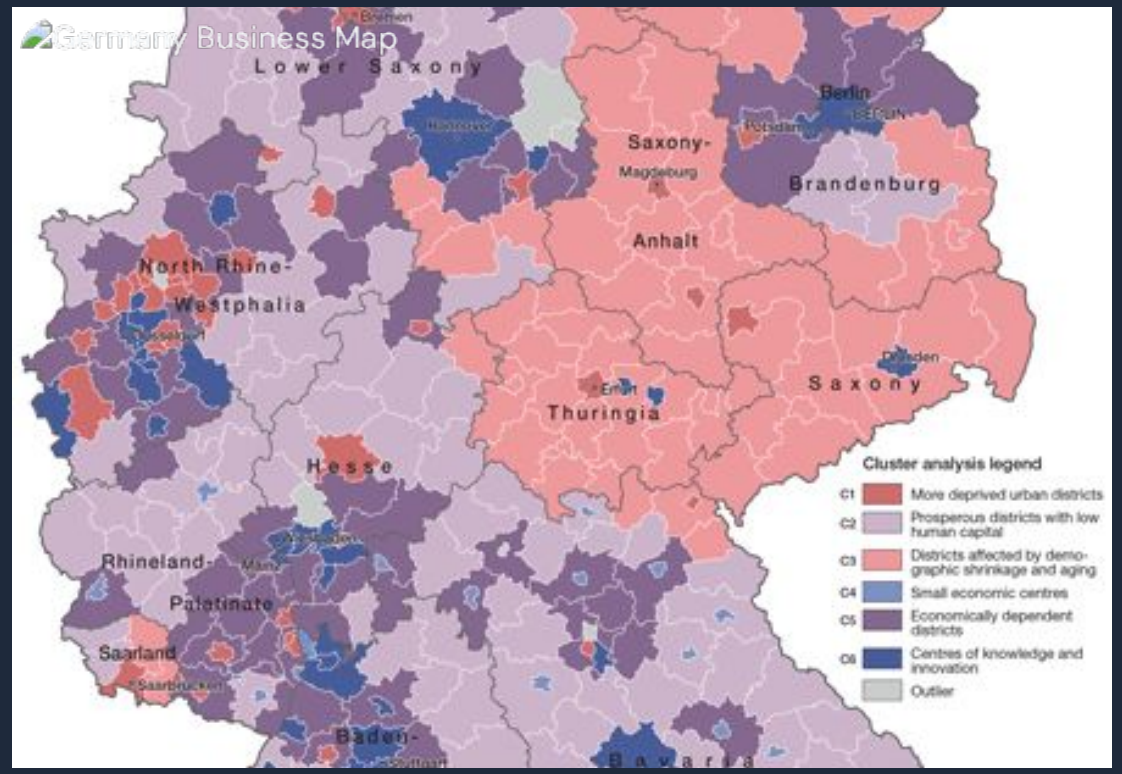
The Germany Crisis

Germany represents our most critical churn risk. Despite having a similar customer base size to Spain, attrition is nearly **double**.

Germany Churn: 32.44%

France/Spain Churn: ~16.0%

Implication: Local market competitive forces or service delivery issues in the German region require immediate localized intervention.



THE DEMOGRAPHIC HIGH-RISK SEGMENTS

Age 45–59

44.04% Churn Rate

Our most financially stable and valuable demographic is exiting at the highest rate.

Age 30–44

42.66% Churn Rate

The "Growth" segment is highly volatile, showing low commitment to legacy banking.

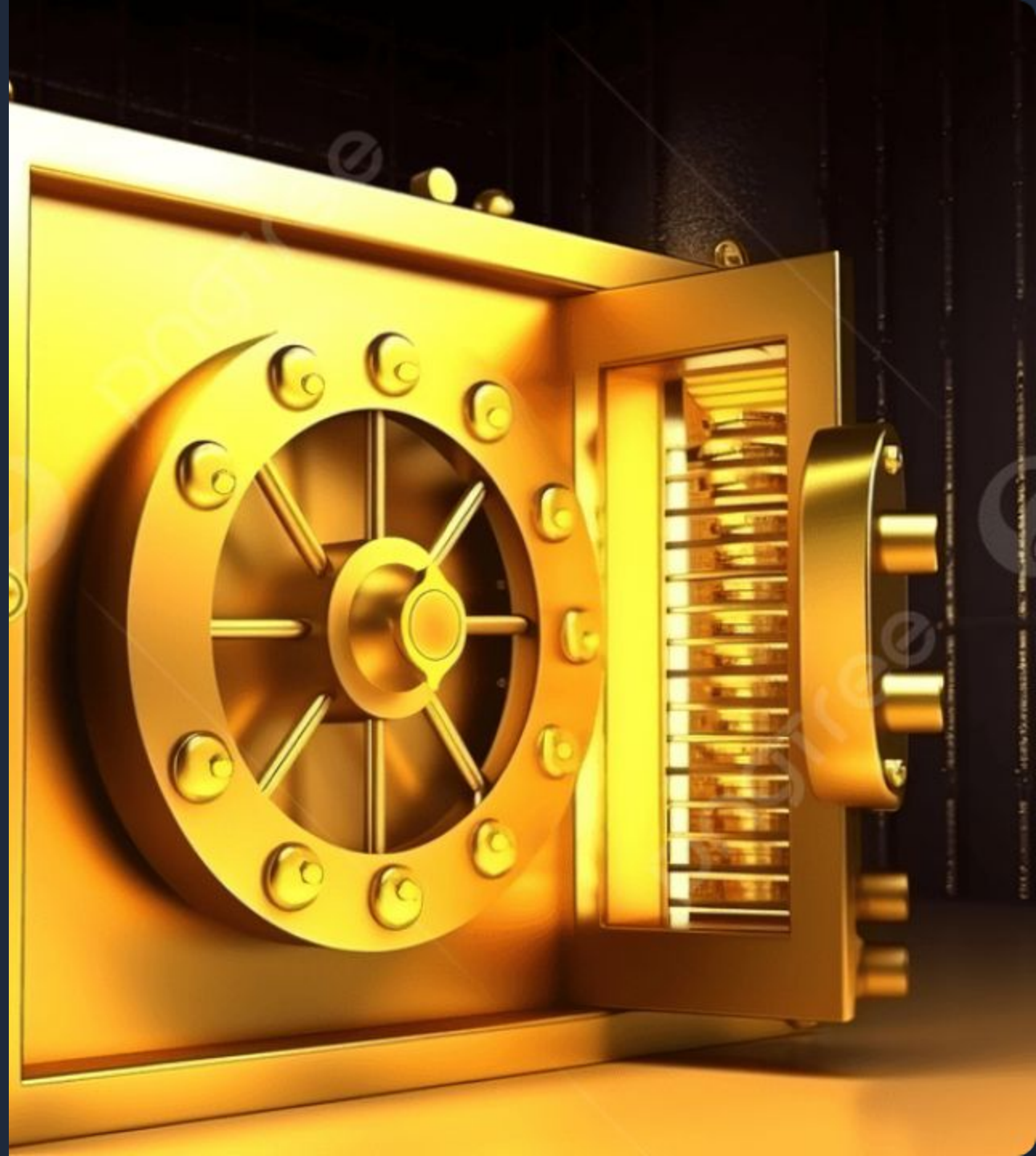
FINANCIAL PARADOX

Leaving With Capital

A staggering **75.45% of churned customers** still had an active balance at the time of exit. This disproves the "forced exit" theory.

These are not "broke" customers. These are affluent clients **consciously deciding** that our value proposition is inferior to a competitor's offer.

Churn is a Service Crisis, not a Credit Crisis.



ENGAGEMENT: THE SILENT KILLER

INACTIVE CHURN	ACTIVE CHURN
27%	14%

Inactivity is the **strongest predictor** of attrition. Customers go "silent" months before moving their capital. We have a window of opportunity during the dormancy phase to intervene.

STRATEGIC ROADMAP

- **Product 2 Conversion:** Shift marketing spend from acquisition to internal cross-selling for 1-product customers.
- **Product 3 & 4 Audit:** Investigate service bottlenecks causing 80%+ churn in high-product households.
- **Early Warning System:** Build CRM alerts for "Dormant" status (0 logins/30 days) to trigger proactive retention offers.